On Languages Describing Large Graph Classes

Henning Fernau[1(✉)], Pamela Fleischmann[2], Kevin Mann[1], and Silas Cato Sacher[1]

[1] Trier University, Trier, Germany
fernau@uni-trier.de , mann@uni-trier.de , sacher@uni-trier.de
[2] Kiel University, Kiel, Germany
fpa@informatik.uni-kiel.de

Abstract. In this work, we introduce a new notion for representing graph classes with formal languages. In contrast to the seminal work by Kitaev and Pyatkin to represent graphs by words, we use formal binary languages in order to have a set of patterns (given by the languages' words) defining the edges in the graph. In particular, we investigate famous languages like the palindromes, copy-words, Lyndon words and Dyck words to represent all graphs or specific graph classes by restricting these languages.

Keywords: Generalized word-representability · Graph classes · Word-representable graphs · Comparability graphs · k-colorable graphs · Palindromes · Dyck-language · Copy-language · Lyndon words

1 Introduction

Graphs are a fundamental theoretical notion for describing relations between anything imaginable. Thus, the question how to store efficiently formally given graphs in computers is as old as computer science itself, and the easiest way is the adjacency matrix which needs quadratic space in the number of the graph's nodes. In 2008 Kitaev and Pyatkin introduced the notion of word-representability [13] and even though it was meant to describe the Perkins semigroup, it also works as an elegant way to store a graph: the graph's nodes are the word's letters and there is an edge between the nodes u and v iff the letters u and v alternate in the representing word, e.g., the *hour-glass graph* on five nodes $V = \{1, 2, 3, 4, 5\}$ where $1, 2, 3$ and $3, 4, 5$ build two triangles can be represented by 124534512. Whereas every graph can be represented by an adjacency matrix, not all graphs are representable by words in this way (cf. the wheel graph W_5 [12,13]). Moreover, it is known to be NP-hard to decide whether a graph is representable or not. Such results motivate research into several directions including the one of altering the definition in order to represent all graphs.

Several variations and generalizations of word-representability have been described in the literature. The alternation of letters a and b in a word's projection onto {a, b} equivalent to the avoidance of the factors aa and bb. This leads to

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
M.-P. Béal and P. Caron (Eds.): DLT 2026, LNCS 16578, pp. 209–222, 2026.
https://doi.org/10.1007/978-3-032-28404-4_16

the generalization via pattern avoiding words in [11]. For any pattern $u \in \{1,2\}^*$ such that $|u| \geq 2$, a graph $G = (V, E)$ is u-representable iff $\text{alph}(w) = V$ and $\{\mathtt{a}, \mathtt{b}\} \in E$ iff w avoids the pattern u (cf. [11, Def. 2] for a formal definition). However, only in the case $u \in \{1\}^*$, their setting can be seen as defining abstract graphs and is hence comparable to classical word-representable graphs. Remarkably, for $|u| \geq 3$ every graph is u-representable. In particular, for $k \geq 3$, every abstract graph is 1^k-representable. The *k-11-representability* generalizes this notion: a graph $G = (V, E)$ is *k-11-represented* by a word $w \in V^*$ if, for all distinct vertices $\mathtt{a}, \mathtt{b} \in V$, at most k times the patterns (factors) $\mathtt{aa}$ and $\mathtt{bb}$ occur in w's projection onto $\{\mathtt{a}, \mathtt{b}\}$. By definition, the 0-11-representable graphs are the classical word-representable graphs, while it can be shown that the 2-11-representable graphs are the class of all graphs [3]. A different approach is the *permutation representability* as defined in [14]: here additional conditions on the words are given that can be used to represent graphs, not only on the patterns that define edges. Lozin [15] discussed several representations of graphs with finite automata. Another completely different way to connect the theory of regular languages with basic notions of graph theory, e.g., with bounded treewidth, was proposed in [5]. In [9] it is shown that the language containing all words representing a specific graph is regular. However, there only one graph is investigated rather than an entire graph class. Yet, there are other ways to connect words to graphs, and hence languages to graph classes. The notion of *letter graphs* introduced by Petkovšek in [16] gives rise to an infinite chain of graph classes where the words are simply restricted by the alphabet's size; in this context, this yields the graph parameter lettericity that recently obtained a certain popularity, cf. [1] and related papers. A generalization of lettericity is contained in [7].

Own Contributions. Our approach seeks to generalize classical word-representability by allowing for a broad variety of patterns in contrast to allowing or avoiding just one. As with classical word-representable graphs, the graph's nodes are the word's letters but there is an edge between the nodes u and v iff the pattern of u and v matches a word from a given binary language L, i.e., $h_{u,v}$ projects the word to $\{u, v\}$, maps u to 0, v to 1 and checks whether the resulting word is in L. For example, $h_{u,v}(uuxvx) = 001$. The language $L_{\text{cl}} \coloneqq (1 \cup \lambda) \cdot (01)^* \cdot (0 \cup \lambda)$ (λ is the empty word) is modeling classical word representability. Moreover, graph representation via pattern avoiding words, yields exactly the graphs characterized by $L_{\neg u} = \overline{\{0,1\}^*\{h_{1,2}(u)\}\{0,1\}^*} \cap \overline{\{0,1\}^*\{h_{2,1}(u)\}\{0,1\}^*}$ for a given pattern $u \in \{1,2\}^*$ with $|u| \geq 2$. Thus, language-representability can be understood as a generalization of graph representation via pattern-avoiding words. In particular, $L_{\neg 1^k}$ describes the class of all graphs for each $k \geq 3$. The k-11-representable graphs can be modeled in our approach as well.

In this work, we study the graph classes described by binary versions of famous languages such as the palindromes, the copy-language, the Dyck-language or Lyndon words. All of those, except for the Dyck-language, represent the class of all graphs and yield a representation that is as good as other implicit graph representations. The graphs described by the Dyck-language describe exactly the

comparability graphs. We also show that we obtain specific graph classes if we intersect the aforementioned languages with other languages, e.g., the language for the classical word-representability.

2 Preliminaries: Fixing General Notions and Notation

In this section we introduce all notations and notions we need for our results and the framework in general. Let $\mathbb{N}$ denote the natural numbers including 0, let $[n] = \{i \in \mathbb{N} \mid 1 \leq i \leq n\}$ and $\mathbb{N}_{\geq n} = \{k \in \mathbb{N} \mid k \geq n\}$ for some $n \in \mathbb{N}$. For a set X and $k \in \mathbb{N}$, define $\binom{X}{k} = \{Y \subseteq X \mid |Y| = k\}$.

Language Theory and Combinatorics on Words. An *alphabet* Σ is a non-empty, finite set with *letters* as elements. A *word* over Σ is a finite *concatenation* of letters from Σ; Σ^* denotes the set of all words over Σ, including the *empty word* λ. The *length of a word* w, i.e., the number of its letters, is denoted by $|w|$. Let $\Sigma^+ = \Sigma^* \setminus \{\lambda\}$. For $w \in \Sigma^*$ and for all $i \in [|w|]$, $w[i]$ denotes the ith letter of w. The number of occurrences of $\mathtt{a} \in \Sigma$ in $w \in \Sigma^*$ is defined as $|w|_{\mathtt{a}} = |\{i \in [|w|] \mid w[i] = \mathtt{a}\}|$ and w's alphabet is given by $\text{alph}(w) = \{\mathtt{a} \in \Sigma \mid \exists i \in [|w|] : w[i] = \mathtt{a}\}$ as the symbols that occur in w. Define $F(w) = \{|w|_a \mid a \in \Sigma\}$ as the *frequentnesses*. A word $w \in \Sigma^*$ is called *k-uniform* for some $k \in \mathbb{N}$ if $|w|_{\mathtt{a}} = k$ for all $\mathtt{a} \in \Sigma$.

Define the *reversal* w^R of $w \in \Sigma^*$ by $w^R = w[|w|] \cdot w[|w| - 1] \cdots w[1]$; w is a *palindrome* if $w = w^R$ holds. The word $w \in \Sigma^*$ is called a *repetition* if there exist $u \in \Sigma^*$ and $k \in \mathbb{N}_{\geq 2}$ such that $w = u^k$ where $u^0 = \lambda$ and $u^k = uu^{k-1}$. Repetitions with $k = 2$ are called *copy-words* and words that are not a repetition are called *primitive*. We also use the *shuffle product* to combine two words: for $u, v \in \Sigma^*$, define the *shuffle* by $u \sqcup\!\sqcup v \coloneqq \{x_1y_1x_2y_2 \cdots x_ny_n \mid \exists x_1, \ldots, x_n, y_1, \ldots, y_n \in \Sigma^* : u = x_1 \cdots x_n \wedge v = y_1 \cdots y_n\}$.

One very famous set of words are the *Lyndon words*. Given a linear ordering $\prec$ on Σ, we extend it to the linear *lexicographical order* on Σ^*, again denoted by $\prec$ and defined by $u \prec w$ iff either u is a *prefix* of w, i.e., $w = ux$ for some $x \in \Sigma^*$, or $w = x\mathtt{b}y_1$, $u = x\mathtt{a}y_2$ for letters $\mathtt{a} \prec \mathtt{b}$ and $x, y_1, y_2 \in \Sigma^*$. The second required basis for Lyndon words is the notion of *conjugates*: the words uv and vu (for $u, v \in \Sigma^*$) are known as *conjugates*. Thus, Σ^* is partitioned in *conjugacy classes*. Finally, a primitive word is a *Lyndon word* if it is the lexicographically smallest element in its conjugacy class. If we want to make the ordering explicit, we also write $\prec$-Lyndon word, or $\prec$-smaller, etc. A mapping f from the free monoid Σ^* into another monoid is called a *morphism* if $f(xy) = f(x)f(y)$ holds for all $x, y \in \Sigma^*$. Note that a morphism is uniquely defined by giving the images of all letters. For $A \subseteq \Sigma$, the *projective morphism* $h_A : \Sigma^* \to A^*$ is defined by $h_A(\mathtt{a}) = \mathtt{a}$ for $\mathtt{a} \in A$ and $h_A(\mathtt{a}) = \lambda$ otherwise. Let $\widetilde{\ }\colon \{0,1\}^* \to \{0,1\}^*$ be the *complement morphism* mapping 0 to 1 and 1 to 0 which is an involution, i.e., $\widetilde{\widetilde{w}} = w$.

Each $L \subseteq \Sigma^*$ is called a *language* over Σ. We extend the concatenation to languages, so that we can define powers of a language L and the *Kleene*

star of L as $L^* = \bigcup_{n\in\mathbb{N}} L^n$ (analogously, we define other operations on words for entire languages). We define the *symmetric hull* operator $\langle L\rangle := L \cup \widetilde{L}$. A language $L \subseteq \{0,1\}^*$ is called 0-1-*symmetric* if $L = \widetilde{L}$ (i.e., $L = \langle L\rangle$). For instance, the language $L_{\rm cl} = (1 \cup \lambda)(01)^*(0 \cup \lambda)$ is 0-1-symmetric. This also holds for its complement $\overline{L_{\rm cl}}$ where $\overline{L}$ denotes $\Sigma^*\backslash L$ for some language $L \subseteq \Sigma^*$. Let $\mathrm{freq}(L) = \{n \in \mathbb{N}_{\geq 1} \mid \exists w \in L : |w|_0 = n\}$ denote the set of frequentnesses in a 0-1-symmetric language $L \subseteq \{0,1\}^*$. The language $\mathcal{D} = \{w \in \{0,1\}^* \mid |w|_0 = |w|_1 \wedge \forall i \in [|w|] : |w[1..i]|_0 \leq |w[1..i]|_1\}$ is the restricted Dyck-language with one pair of parentheses with 0 for the opening and 1 for the closing bracket (cf. $D_1'^*$ in [2]). More generally, $D_k'^*$ is the set of correctly parenthesized words formed with k different types of parentheses, forming an alphabet Σ of size $2k$. Identifying singleton sets with their elements, we can build *regular expressions* from letters by using concatenation $\cdot$, union $\cup$, and Kleene star. We also include set complementation $\bar{\cdot}$ and the *shuffle* $\sqcup\!\sqcup$ when building such expressions.

Graph Theory. Throughout this paper, we only consider undirected finite graphs. A graph G is a pair (V, E) with the finite, non-empty set of vertices V and the set of edges $E \subseteq \binom{V}{2}$. For a given graph G, let $V(G)$ and $E(G)$ denote its sets of vertices and edges respectively. The cardinality of V is called the *order* of G. If $\{u, v\} \in E$, u is called a *neighbor* of v, and $N(v) \subseteq V$ are the neighbors of v; $N[v] := N(v) \cup \{v\}$ is the *closed neighborhood* of v. The *degree* of $v \in V$ is $\deg(v) := |N(v)|$. The *complement* of the graph G, written $\overline{G}$, satisfies $\overline{G} = (V, \binom{V}{2} \setminus E)$. For a graph class $\mathcal{G}$, co-$\mathcal{G} := \{\overline{G} \mid G \in \mathcal{G}\}$. If $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are graphs, then $\varphi : V_1 \to V_2$ is a *graph morphism* iff $\{u, v\} \in E_1$ implies $\{\varphi(u), \varphi(v)\} \in E_2$; a bijection φ is a *graph isomorphism* iff $\{u, v\} \in E_1 \iff \{\varphi(u), \varphi(v)\} \in E_2$; then we write $G_1 \simeq G_2$. We call a graph with $n \in \mathbb{N}$ vertices and an empty edge set a *null graph*, denoted by N_n, and a graph with n vertices and all possible edges a *complete graph*, denoted by K_n, i.e. $\overline{N_n} = K_n$. A graph with $n + m$ vertices, with $n, m \in \mathbb{N}_{\geq 1}$ is called a *complete bipartite graph* $K_{n,m} = (V, E)$ if $V = U \cup W$, $U \cap W = \emptyset$, with $|U| = n$ and $|W| = m$ and E contains all edges between U and W but no other edges. A graph with $n \in \mathbb{N}$ vertices is a *path* $P_n = (V, E)$ if there is a linear ordering $<$ on its vertex set such that $\{u, v\} \in E$ if either u is the immediate predecessor of v in $<$ or u is the immediate successor of v in $<$. A graph with $n \in \mathbb{N}$ vertices is a *cycle* $C_n = (V, E)$ if it contains a path $P_n = (V, E')$ and one additional edge e that connects the two vertices of this path that have degree one. A vertex in a graph is called *isolated* if $N(v) = \emptyset$. If $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are two graphs with disjoint sets of vertices V_1 and V_2, then the (graph) *union* of G_1 and G_2 is $G_1 \cup G_2 = (V_1 \cup V_2, E_1 \cup E_2)$. Let $G = (V, E)$ be a graph with a partition $V = V_1 \cup V_2$. If $\binom{V_1}{2} \cap E = \emptyset$ and $\binom{V_2}{2} \cap E = \emptyset$, then G is called *bipartite*. If $\binom{V_1}{2} \cap E = \emptyset$ and $\binom{V_2}{2} \cap E = \binom{V_2}{2}$, then G is called a *split graph*. If $\binom{V_1}{2} \cap E = \binom{V_1}{2}$ and $\binom{V_2}{2} \cap E = \binom{V_2}{2}$, then G is called *cobipartite*. k-colorability generalizes bipartiteness by partitioning the vertex set into k independent sets instead of two. If there exists a partial order on the vertex set V of $G = (V, E)$

such that $u, v \in V$ are incomparable if and only if $\{u, v\} \in E$, then G is a *comparability graph*.

Our Main Notions. After establishing the basics from language and graph theory, we are now able to introduce the main notions for our setting. We start with a definition, which allows us to compare words over an alphabet of a graph's nodes with a binary language. This new morphism $h_{\mathtt{a},\mathtt{b}}$ can be viewed as the composition of the projection $h_{\{u,v\}}$ and the renaming isomorphism $\iota : \{u, v\}^* \to \{0, 1\}^*$ with $u \mapsto 0$ and $v \mapsto 1$ for some given nodes u, v. For example, if we have $\mathtt{a}, \mathtt{b}, \mathtt{n} \in V$ and $\mathtt{banana} \in V^*$, we get $h_{\mathtt{a},\mathtt{b}}(\mathtt{banana}) = 1000$. If we have 0-1-symmetric languages, it is unimportant whether u or v is mapped to 0 or 1 resp.; it only influences later the names of the represented nodes.

Definition 1. *Let V be an alphabet and fix $u, v \in V$ with $u \neq v$. Define $h_{u,v} : V^* \to \{0, 1\}^*$ by $u \mapsto 0$, $v \mapsto 1$ and $x \mapsto \lambda$ for $x \in V \setminus \{u, v\}$.*

Based on $h_{u,v}$, we introduce the notion of L-representability of graphs.

Definition 2. *Let $L \subseteq \{0, 1\}^*$ be 0-1-symmetric. Let $w \in V^*$ with $V = \mathrm{alph}(w)$. We call a graph $G = (V, E)$ L-*represented by $w \in V^*$ *iff, for all $u, v \in V$ with $u \neq v$, we have $\{u, v\} \in E \Leftrightarrow h_{u,v}(w) \in L$. A graph $G = (V, E)$ is L-*representable *if it can be L-represented by some word $w \in V^*$. Let $G(L, w)$ denote the graph G that is L-represented by $w \in \Sigma^*$ and let $\mathcal{G}_L = \{G(L, w) \mid w \in \Sigma^*, \Sigma = \mathrm{alph}(w)\}$.*

As 0-1-symmetry is essential for $G(L, w)$ and $\mathcal{G}_L$ being well-defined, we will assume this condition for any binary language used for defining graphs and graph classes in the following. We finish this section with some immediate insights and examples. Since $h_{u,v} \in L$ iff $h_{u,v} \notin \overline{L}$ for every language L and vertices u, v, we have $G(L, w) = \overline{G(\overline{L}, w)}$ and we get:

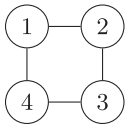
Lemma 3. *For each $L \subseteq \{0, 1\}^*, \mathcal{G}_{\overline{L}} =$ co-$\mathcal{G}_L$.*

Lemma 4. *Let V be an alphabet and $w \in V^*$. If $0^k \sqcup\!\sqcup\, 1^\ell \subseteq L \subseteq \{0, 1\}^*$ for all $k, \ell \in F(w)$, then $K_{|V|}$ is L-represented by w.*

Lemma 5. *A graph G can be L-represented iff it can be L^R-represented.*

Corollary 6. *If $L \subseteq \{0, 1\}^*$ satisfies $L = L^R$, then $G(L, w) = G(L, w^R)$ for each word w.*

Intuitively, one may assume that G is L-representable iff G is $(L \cup L^R)$-representable but consider given $L = \overline{\langle 0001 \rangle}$ and $w = \mathtt{aaab}$, we have $G(L, w) = (\{\mathtt{a}, \mathtt{b}\}, \emptyset)$, $L \cup L^R = \{0, 1\}^*$, and $\mathcal{G}_{L \cup L^R}$ is the class of complete graphs and $G \notin \mathcal{G}_{L \cup L^R}$. We finish this section by briefly looking into 0-1-symmetric languages L that are closed under reversal ($L = L^R$), e.g., L_{cl} is closed under reversal. Note that L_{cl} equals $L_{\neg 1^k} = \overline{\{0, 1\}^* \{0^k\} \{0, 1\}^* \cap \{0, 1\}^* \{1^k\} \{0, 1\}^*}$ for $k = 2$. By [11], we know that $\mathcal{G}_{L_{\neg 1^k}}$ corresponds to the family of graphs called 1^k-representable. There it is also shown that, for every $k \geq 3$, *every* graph is 1^k-representable., i.e., $\mathcal{G}_{L_{\neg 1^k}}$ is the class of all (undirected) graphs. For instance, we have for $k = 3$, $G(L_{\neg 1^3}, w) = ([4], \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}, \{2, 4\}\})$ with the word $w = 1124112341234$. More representations of the cycle C_4 can be found in Fig. 1.



L	w
$L_{\neg 1^3}$ [11, Thm. 4]	2213212431124112341234
L_{1112} [10, Thm. 3.12]	2213241243112413123412341
$L_{2\text{-}11}$ [3, Thm. 5.2]	1432341241322431423124314312431 2
L_{P} (Thm. 9)	423121123142
$L_{\mathcal{C}}$ (Thm. 15)	121324123142
$L_{\mathcal{L}}$ (Thm. 24)	11122233344412341234112411323423422322434343343344444 4
$L_{\mathcal{D}}$ (Thm. 27)	12341234132412341234

Fig. 1. Language L represents C_4 with the word w.

3 Representing Graphs with Famous Languages

In this section, we turn our attention to rather 'famous' (infinite, non-regular) languages as the set of palindromes, the-copy language, the Dyck-language or the set of Lyndon words and look at their associated graph classes. Thus, in all paragraphs we start with the associated binary 0-1-symmetric languages. We will see that most of these languages over the arbitrary alphabets allow to describe all graphs in a rather efficient way. Illustrations of these representations and the underlying constructions can be found in Fig. 1. Some proofs were omitted due to space restrictions; this is marked by $(*)$.

Palindromes. Set $L_{\mathcal{P}} := \{u \in \{0,1\}^+ \mid u = u^R\}$. This language can be restricted to the deterministic context-free version of the binary palindrome language $L_{\text{detP}} := \{h_{\text{double}}(w)01(h_{\text{double}}(w))^R, h_{\text{double}}(w)10(h_{\text{double}}(w))^R \mid w \in \{0,1\}^*\}$ with the morphism $h_{\text{double}} : \{0,1\} \to \{0,1\}$ such that $h(\mathtt{a}) = \mathtt{aa}$ for all $\mathtt{a} \in \{0,1\}$. Motivated by [9], we also define $\mathcal{G}_\star = \{K_{1,n} \cup N_m \mid n, m \in \mathbb{N}\}$ as the set of all extended star graphs. Regarding palindromes in general, notice that $h_{\mathtt{a},\mathtt{b}}(w^R) = h_{\mathtt{a},\mathtt{b}}(w)^R$ for each palindrome $w \in \Sigma^*$ and all $\mathtt{a}, \mathtt{b} \in \Sigma$. Our first lemma shows that being a palindrome can be characterized by the palindrome property of all projections to binary sub-alphabets. Besides being a nice word-combinatorial property, it has a graph-theoretic interpretation, see Corollary 8.

Lemma 7. $(*)$ *A word $w \in \Sigma^*$ is a palindrome iff, for all $\mathtt{a}, \mathtt{b} \in \Sigma$ with $\mathtt{a} \neq \mathtt{b}$, $h_{\mathtt{a},\mathtt{b}}(w)$ is a palindrome.*

The previous lemma links the set of all palindromes to $L_{\mathcal{P}}$ and we show that the (deterministic) palindrome languages represents all graphs.

Corollary 8. *A word w is a palindrome iff $G(L_{\mathcal{P}}, w) \simeq K_{|\operatorname{alph}(w)|}$.*

Theorem 9. *The binary palindrome language $L_{\mathcal{P}}$ represent every graph.*

Proof. Let $G = ([n], E)$ be a graph with $n \in \mathbb{N}$. Define, for $i \in [n]$, $u_i \in [i-1]^*$ as the word which enumerates $\overline{N(i)} \cap [i-1]$ in linear order as well as $w_1 := 11$ and $w_i := iu_iw_{i-1}iu_i^R$ if $i \in \mathbb{N}_{\geq 2}$. We want to show that $h_{j,k}(w_i) = h_{j,k}(w_i)^R$ iff $\{j,k\} \in E$ for all $i \in [n]$ and $j, k \in [i]$ with $j \neq k$. For $i = 1$ this

trivially holds. So let $i > 1$. Since $h_{j,k}$ is a morphism, $h_{j,k}(iu_iw_{i-1}iu_i^R)^R = h_{j,k}(u_i^R)^R h_{j,k}(i)^R h_{j,k}(w_{i-1})^R h_{j,k}(u_i^R) h_{j,k}(i)^R = h_{j,k}(u_i i w_{i-1}^R u_i^R i)$.

First we assume $i \notin \{j,k\}$. Then $h_{j,k}(w_i) = h_{j,k}(u_i)h_{j,k}(w_{i-1})h_{j,k}(u_i^R)$. This equals $h_{j,k}(w_i^R) = h_{j,k}(u_i)h_{j,k}(w_{i-1})^R h_{j,k}(u_i^R)$ iff $h_{j,k}(w_{i-1})^R = h_{j,k}(w_{i-1})$. By induction, we know this is the case iff $\{j,k\} \in E$. Now we can assume $j = i$. If $\{i,k\} \notin E$, then $h_{i,k}(w_i) = h_{i,k}(u_i) \cdot h_{i,k}(i) \cdot h_{i,k}(w_{i-1}) \cdot h_{i,k}(u_i^R) \cdot h_{i,k}(i) = 1 \cdot 0 \cdot h_{i,k}(w_{i-1}) \cdot 1 \cdot 0$ as $k < i$, which is no palindrome. For $\{i,k\} \in E$, $h_{i,k}(w_i) = 0h_{i,k}(w_{i-1})0$. This is a palindrome as $h_{i,k}(w_{i-1}) \in 1^+$. By induction, $G \simeq G(L_{\mathcal{P}}, w_n)$ follows. □

Theorem 10. *$\mathcal{G}_{L_{\mathrm{detP}}}$ is the class of all graphs.*

Proof. Let $G = ([n], E)$ be a graph with $n \in \mathbb{N}$. Define $\langle M \rangle$ as the words that enumerate the vertices of the set $M \subseteq [n]$ in linear order. Define, for $i \in [n]$, $u_i := h_{\mathrm{double}}(\langle \overline{N(i)} \cap [i-1] \rangle)$ and

$w_i := 11\langle [n] \rangle$ if $i = 1$ and $w_i := iiu_iw_{i-1}iiu_i^R$ if $i \neq 1$. We show $G(L_{\mathrm{detP}}, w_n)$ analogous to the proof of Theorem 9. □

By [9, Theorem 28], the non-empty palindromes represent exactly the extended star graphs in the following sense: $\mathcal{G}_\star = \{G(L_{\mathrm{cl}}, w) \mid w \text{ is a palindrome}\}$. This raises the question if we can combine the classical word-representability with our setting in the sense of intersecting $L_{\mathcal{P}}$ and L_{cl} to obtain the class of extended star graphs. The following theorem shows that in fact, we have $\mathcal{G}_\star \subsetneq \mathcal{G}_{L_{\mathcal{P}} \cap L_{\mathrm{cl}}}$, since the latter one is the class of bipartite graphs.

Theorem 11. $(*)$ *$\mathcal{G}_{L_{\mathcal{P}} \cap L_{\mathrm{cl}}}$ is the class of bipartite graphs.*

Since Theorem 11 shows that intersecting the palindrome language with the language for the classical word-representability may lead to interesting (even on first sight unexpected) results, we finish the paragraph about palindromes with two other combinations of languages.

Corollary 12. $(*)$ *1. $\mathcal{G}_{(L_{\mathcal{P}} \cap L_{\mathrm{cl}}) \cup ((0^2)^+ \sqcup (1^2)^+)}$ is the class of split graphs.*
2. $\mathcal{G}_{\overline{L_{\mathcal{P}}} \cup \overline{L_{\mathrm{cl}}}}$ is the class of cobipartite graphs.

Copy-Language. Let $L_{\mathcal{C}} = \{ww \mid w \in \{0,1\}^*\}$ denote the binary copy-language which is 0-1-symmetric. Apart from the fact that the language of palindromes is context-free, while the copy-language is not, both languages enjoy quite similar properties. We exemplify this by an analogue to Lemma 7. We follow a presentation similar to the one on palindromes, starting with a morphic characterization.

Lemma 13. $(*)$ *For every alphabet Σ such that $|\Sigma| \geq 2$, a word $w \in \Sigma^*$ is a copy-word iff, for all $\mathtt{a}, \mathtt{b} \in \Sigma$ with $\mathtt{a} \neq \mathtt{b}$, $h_{\mathtt{a},\mathtt{b}}(w)$ is a copy-word.*

Similarly to the palindromes, we can infer from Lemma 13 that we again obtain the complete graphs and in general that $_{\mathcal{C}}$ represents all words.

Corollary 14. *A word w is a copy-word iff $G(L_{\mathcal{C}}, w) \simeq K_{|\operatorname{alph}(w)|}$.*

Theorem 15. *The binary copy-language $L_{\mathcal{C}}$ represent every graph.*

Proof. Let $G = ([n], E)$ be a graph. Define $w = u_1 1 u_2 2 \cdots u_n n 1 u_1 \cdots n u_n$ where u_i enumerates each vertex of $\overline{N(i)} \cap [i]$ exactly once in the natural order for $i \in [n]$. Consider $G' = ([n], E') = G(L_{\mathcal{C}}, w)$. Let $i, j \in [n]$ with $i < j$. By definition, $h_{i,j}(u_i) = \lambda$. Also, $h_{i,j}(u_j) \in \{\lambda, 0\}$ with $h_{i,j}(u_j) = \lambda$ being equivalent to $i \in N(j)$. For $k < i$, $h_{i,j}(u_k) = \lambda$. Hence, $h_{i,j}(w) = w_1 w_2$ with

$$w_1 \coloneqq h_{i,j}(u_i) 0 h_{i,j}(u_{i+1}) \cdots h_{i,j}(u_j) 1 h_{i,j}(u_{j+1}) \cdots h_{i,j}(u_n) \text{ and}$$
$$w_2 \coloneqq 0 h_{i,j}(u_i) h_{i,j}(u_{i+1}) \cdots 1 h_{i,j}(u_j) h_{i,j}(u_{j+1}) \cdots h_{i,j}(u_n) .$$

It is easy to see that $|w_1| = |w_2|$. Furthermore, $w_1 = w_2$ holds iff $1 h_{i,j}(u_j) = h_{i,j}(u_j) 1$. Since $h_{i,j}(u_j) \in \{\lambda, 0\}$, $1 h_{i,j}(u_j) = h_{i,j}(u_j) 1$ iff $h_{i,j}(u_j) = \lambda$. As $h_{i,j}(u_j) = \lambda$ iff $i \in N(j)$, we can conclude $E = E'$ and $G = G'$. □

Estimating the complexity of a language by the Chomsky hierarchy, the copy-language is quite complicated. Notice that by Lemma 3, we get another characterization of all graphs through complementation, and the complement of the copy-language is a one-counter language, a subclass of context-free.

Corollary 16. *The complement of $L_{\mathcal{C}}$ can represent every graph.*

Now we intersect $L_{\mathcal{C}}$ with the the classical word-representation. Moreover, we generalise this result: copy-words are only a special case of repetitions w^i for some $i \in \mathbb{N}_{\geq 2}$.

Theorem 17. (∗) *$\mathcal{G}_{L_{\mathcal{C}} \cap L_{\mathrm{cl}}}$ is the set of classically word-representable graphs.*

Lemma 18. (∗) *Let $w \in \Sigma^*$ and $i \in \mathbb{N}_{\geq 1}$. Then w^i and w L_{cl}-represent the same graph iff for all $\mathtt{a}, \mathtt{b} \in \Sigma$ we have $h_{\{\mathtt{a},\mathtt{b}\}}(w) \in \mathtt{a}\{\mathtt{a},\mathtt{b}\}^*\mathtt{b} \cup \mathtt{b}\{\mathtt{a},\mathtt{b}\}^*\mathtt{a}$.*

Proposition 19. (∗) *Let $L = \langle 0\{0,1\}^*1 \rangle$. If w L-represents G, then also w^i L-represents G for all $i \in \mathbb{N}_{\geq 1}$.*

We finish this paragraph with an – in our opinion – unexpected twist w.r.t. restricting the copy-language $L_{\mathcal{C}}$. Define for all $k \in \mathbb{N}$, the restriction $L_{\mathcal{C},k} \coloneqq \{ww \mid |w|_0 \not\equiv_k |w|_1\}$ of $L_{\mathcal{C}}$ where the number of 0s and 1 resp. is not allowed to be congruent modulo k. This language represents exactly the k-colorable graphs.

Theorem 20. (∗) *$\mathcal{G}_{L_{C,k}}$ is the class of k-colorable graphs.*

The construction of the words representing $G = ([n], E)$ is similar to the idea of the copy-language: Let $f : V \to \{1, \ldots, k\}$ be the coloring. We use $x = 1^{f(1)} 2^{f(2)} \cdots n^{f(n)}$ and $w = x u_1 1^k u_2 2^k \cdots u_n n^k x 1^k u_1 \cdots n^k u_n$ where u_i writes t^k exactly once per $t \in \overline{N(i)} \cap [i]$ times in the natural order. Observe that x ensures the coloring. For all $i \in [n]$, $u_i i^k$ and $i^k u_i$ deletes the edges between i and the elements in $\overline{N(i)} \cap [i]$. This construction can be generalized to graph morphisms.

Theorem 21. (*) *For all graphs $H = (U, F)$, there is a language $L(H)$ such that $\mathcal{G}_{L(H)}$ is the set of all graphs $G = (V, E)$ for which there exists a graph homomorphism $f : V \to U$.*

Lyndon Words. Proposition 19 establishes that repetitions of a representant are not needed and even though this word is not necessarily primitive, this insight encourages to look at the Lyndon words. By [13, Prop. 5] we have that if a k-uniform word w represents a graph G in the classical sense, then every word in the conjugacy class of w also represents G, i.e., by [13, Thm. 7], for representing any word-representable graph, Lyndon words suffice. As we focus on 0-1-symmetric languages and as not both $\mathtt{ab}$ and $\mathtt{ba}$ can be Lyndon for any two different $\mathtt{a}, \mathtt{b} \in \Sigma$, we look at the symmetric closure of binary Lyndon words: set $L_{\mathcal{L}} := \langle L_{\mathrm{Lyndon},\{0,1\},<} \rangle$ with the ordering $0 < 1$. Observe that $L_{\mathcal{L}} = L_{\mathrm{Lyndon},\{0,1\},<} \cup L_{\mathrm{Lyndon},\{0,1\}>}$. The results for the copy-language imply that repetitions cannot occur in projections when dealing with Lyndon words. Therefore, we can, w.l.o.g., focus on the property of being the smallest conjugate in its class during the investigation of Lyndon words. N ot all renaming projections $h_{\mathtt{a},\mathtt{b}}(w)$ of a Lyndon word w have to be Lyndon: we have for the Lyndon word $w = \mathtt{abca}$ that $h_{\mathtt{a},\mathtt{b}}(w) = 010$, is not Lyndon. Hence, we only get:

Lemma 22. (*) *Let $w \in \Sigma^*$ such that $h_{\mathtt{a},\mathtt{b}}(w)$ is $<$-Lyndon for all $\mathtt{a}, \mathtt{b} \in \Sigma$ with $\mathtt{a} \prec \mathtt{b}$. Then w is $\prec$-Lyndon.*

Lemma 23. (*) *Let $\mathtt{a}, \mathtt{b} \in \Sigma$ with $\mathtt{a} \neq \mathtt{b}$ and $w \in \Sigma^*$.*
1. $h_{\mathtt{a},\mathtt{b}}(w)$ is a $<$-Lyndon word iff $h_{\mathtt{b},\mathtt{a}}(w)$ is a $>$-Lyndon word.
2. If $h_{\mathtt{a},\mathtt{b}}(w)$ is a $<$-Lyndon word then $h_{\mathtt{a},\mathtt{b}}(w)$ is no $>$-Lyndon word.

Theorem 24. *$\mathcal{G}_{L_{\mathcal{L}}}$ includes every graph.*

Proof. Let $G = ([n], E)$ be an graph with $n \in \mathbb{N}$. Define for $i \in [n]$ the words $v_i := i \cdots n$, $x_i := j_{i,1} \cdots j_{i,\ell_i}$, $y_i := k_{i,1} \cdots k_{i,\ell'_i}$ and $u_i := i^2 x_i i^2 y_i$ where $N(i) \cap [i..n] = \{j_{i,1}, \ldots, j_{i,\ell_i}\}$, $\overline{N(i)} \cap [i..n] = \{i, k_1, \ldots, k_{i,\ell'_i}\}$ and $j_{i,1}, \ldots, j_{i,\ell_i}$ as well as $k_{i,1}, \ldots, k_{i,\ell'_i}$ are strictly monotone increasing. We will show $G = G' = (V, E') := G(L_{\mathcal{L}}, w)$, $w := 1^3 2^3 \cdots n^3 v_1^2 u_1 v_2^2 u_2 \cdots v_n^2 u_n$.

Let $p, q, r \in [n]$ with $p < q$. By Lemma 23 only $h_{p,q}(w), h_{q,p}(w)$ with respect to $0 < 1$ needs to be considered. Since p appears before q in w, it is enough to consider $h_{p,q}(w)$. Observe that for $q < r$, $h_{p,q}(v_r u_r) = \lambda$. Hence, $h_{p,q}(v_{q+1} u_{q+1} \cdots v_n^2 u_n) = \lambda$. For $p < r < q$, $h_{p,q}(v_r^2 u_r) = 111$ and $h_{p,q}(v_q^2 u_q) = 111111$. Thus, $h_{p,q}(v_q u_q \cdots v_n^2 u_n) \in 111111(111)^*$.

Next, assume $r < p < q$. Then $h_{p,q}(v_i) = 01$. If $p, q \in N(r)$, $h_{p,q}(x_r) = 01$ and $h_{p,q}(y_r) = \lambda$. Analogously, $h_{p,q}(x_r) = \lambda$ and $h_{p,q}(y_r) = 01$ for $p, q \notin N(r)$. In the case of $p \in N(r)$ and $q \notin N(r)$, $h_{p,q}(x_r) = 0$ and $h_{p,q}(y_r) = 1$. For $p \notin N(r)$ and $q \in N(r)$, $h_{p,q}(x_r) = 1$ and $h_{p,q}(y_r) = 0$. Therefore, $h_{p,q}(v_r v_r u_r) \in \{010101, 010110\}$ and $h_{p,q}(v_1^2 u_1 \cdots v_{p-1}^2 u_{p-1}) \in \{010101, 010110\}^{p-1}$. Clearly, there does not exist an $i \in [6(p-1)]$ with $w[3n + i..3n + i + 2] = 000$. Finally, consider $r = p$. If $\{p, q\} \in E$ then $h_{p,q}(v_p^2, u_q) = 010100100$ and $h_{p,q}(w) \in$

$000111\{010101, 010110\}^{p-1}010100100111111(111)^*$. Clearly, $i = 1$ is the only $i \in |h_{p,q}(w)|$ such that $h_{p,q}(w)[i..i+2] = 000$. Since $h_{p,q}(w)[|h_{p,q}(w)|] = 1$, $h_{p,q}(w)$ is Lyndon and $\{p, q\} \in E'$.

Assume $\{p, q\} \notin E$. Then $h_{p,q}(v_p p^2 x_p p^2 y_p) = 0100001$. As $v_p u_p = v_p p^2 x_p p^2 y_p$ is an infix of w, 0100001 is an infix of $h_{p,q}(w)$. As 000111 is a prefix of $h_{p,q}(w)$, $h_{p,q}(w)$ is no Lyndon word (a conjugate with prefix $h_{p,q}(u_p)$ is $<$-smaller). Thus, $\{p, q\} \notin E$. In total, $G = G'$. □

We conclude this paragraph with some insights about graph classes when we only look at odd or even length Lyndon words with a similar result to Corollary 12.

Theorem 25. $(*)$ $\mathcal{G}_{\{w \in L_{\mathcal{L}} \mid |w| \text{ is odd}\}}$ *is the class of bipartite graphs.*

Corollary 26. *Let* $L := \{w \in L_{\mathcal{L}} \mid |w| \text{ is odd}\}$, $L_{odd} := \{w \in \{0,1\}^* \mid |w|_0, |w|_1 \text{ are odd}\}$ *and* $L_{even} := \{w \in \{0,1\}^* \mid |w|_0, |w|_1 \text{ are even}\}$. *Then* $\mathcal{G}_{L \cup L_{odd}}$ *is the class of split graphs and* $\mathcal{G}_{L \cup L_{odd} \cup L_{even}}$ *is the class of cobipartite graphs.*

Dyck-language. Similar to the Lyndon words, we need here to define the symmetric closure of the Dyck-language as $L_{\mathcal{D}} := \langle \mathcal{D} \rangle$. Also similar, we have $h_{(,)}(w) \in L_{\mathcal{D}}$ for any pair of parentheses $(,) \in \Sigma$ if $w \in D_k'^*$, but the converse is not true, as $w = ([])$ shows. Our main result is that $\mathcal{G}_{L_{\mathcal{D}}}$ is the class of comparability graphs. To see that each graph in $\mathcal{G}_{L_{\mathcal{D}}}$ is a comparability graph, consider the order $\prec_V$ such that, for all $v, u \in V$, $v \prec_V u$ iff $v \neq u$, $|w|_v = |w|_u$ and, for all prefixes w' of w, $|w'|_v \leq |w'|_u$. For the other direction, let $G = (V, E)$ be a comparability graph with the order $\prec$ on V. For each $v \in V$, $z_v := y_v v x_v$ where x_v enumerates all $u \in V$ with $v \prec u$, and y_v the remaining vertices. For all $v, u \in V$, if $v \prec u$ then $h_{v,u}(z_v) = 01$. The final word enumerates first all vertices and then all z_v in the order of $\prec$.

Theorem 27. $\mathcal{G}_{L_{\mathcal{D}}}$ *is the class of comparability graphs.*

Proof. Let $G = (V, E)$ be a comparability graph w.r.t. the strict order $\prec$ on V, i.e., $\{u, v\} \in E$ iff u, v are comparable with respect to $\prec$. Define the upper set $U_{\prec}(v) := \{u \in V \mid v \prec u\}$ for $v \in V$. $\prec_{lin}$ denotes an arbitrary but fixed linear order on V that extends $\prec$. Define $z := v_1 \cdots v_n$ to be an enumeration of V in the order $\prec_{lin}$. For $v \in V$, let x_v, y_v be words such that x_v enumerates $U_{\prec}(v)$ and y_v enumerates $\overline{U_{\prec}(v)} \setminus \{v\}$ according to the order $\prec_{lin}$ (increasingly). Further, denote $z_v := y_v v x_v$, i.e., z_v first enumerates the elements of V that are in the upper set of v, then v, and then the remaining elements of V. Define $w := z z_{v_1} \cdots z_{v_n}$ and $G' = (V, E') = G(L_{\mathcal{D}}, w)$. Note that for each $v, u \in V$, $|z_v|_u = |z|_u = 1$. Thus, for all $v, a, b \in V$, $h_{a,b}(z_v), h_{a,b}(z) \in \{01, 10\}$. Let w.l.o.g. $a, b \in V$ with $a \prec_{lin} b$ in our following discussion. Thus, $h_{a,b}(z) = 01$ $(*)$.

Assume $\{a, b\} \in E$. As G is a comparability graph, this implies $a \prec b$ $[*]$, as $b \not\prec a$ because $a \prec_{lin} b$. We show that $h_{a,b}(z_v) = 01$ for all $v \in V$. Then, with $(*)$, we can conclude $h_{a,b}(w) = (01)^{n+1}$. Hence, $h_{a,b}(w) \in L_{\mathcal{D}}$. Suppose there existed

$v \in V$ such that $h_{a,b}(z_v) = 10$. Hence, b precedes a in the enumeration z_v. If $v = a$ (or $v = b$, resp.), then $b \in \text{alph}(y_v) = \overline{U_\prec(v)} \setminus \{v\}$ (or $a \in \text{alph}(x_v) = U_\prec(v)$, resp.), contradicting [$*$]. Therefore, $v \notin \{a, b\}$. Since x_v, y_v enumerate in accordance with $\prec_{lin}$ and as $a \prec b$, b has to appear in y_v and a in x_v. This is a contradiction to the transitivity of $\prec$, as $a \prec b$ and $v \prec a$ (because $a \in U_\prec(v)$) would imply $b \in U_\prec(v) = \text{alph}(x_v)$. Hence, $h_{a,b}(z_v) = 01$ for all $v \in V$. Now assume $\{a, b\} \notin E$. Then $|y_a|_b = 1$ and $h_{a,b}(z_a) = 10$. Let $a = v_i$. This implies $h_{a,b}(w) \in 01\{01, 10\}^{i-1}10\{01, 10\}^{n-i}$. Then $\{a, b\} \notin E'$, since $|h_{a,b}(w[1..2i + 1])|_0 = i < i + 1 = |h_{a,b}(w[1..2i + 1])|_1$ and $|h_{a,b}(w[1])|_1 = 0 < 1 = |h_{a,b}(w[1])|_0$.

Conversely, let $G = (V, E) \in \mathcal{G}_{L_\mathcal{D}}$. There exists $w \in V^*$ such that $G = G(L_\mathcal{D}, w)$. Define the order $\prec_V$ such that, for all $v, u \in V$, $v \prec_V u$ iff $v \neq u$, $|w|_v = |w|_u$ and, for all prefixes w' of w, $|w'|_v \leq |w'|_u$. We show that $v \prec_V u$ is a strict partial order and that $\{v, u\} \in E$ iff $v \prec_V u$, i.e., G is a comparability graph. By definition, $\prec_V$ is irreflexive. Let $a, b, c \in V$ with $a \prec_V b$ and $b \prec_V c$. Thus, $|w|_a = |w|_b = |w|_c$. For any prefix w' of w, $|w|_a \leq |w|_b \leq |w|_c$. Let $i \in [|w|]$ be the the smallest i with $w[i] = b$. This implies $|w[1..i - 1]|_a \leq 0 = |w[1..i - 1]|_b$ and $|w[1..i]|_a \leq 0 < 1 = |w[1..i]|_b \leq |w[1..i]|_c$. Thus, $a \neq c$ and $a \prec_V c$. Therefore, $\prec_V$ is a strict partial order.

Let $a, b \in V$. For $|w|_a \neq |w|_b$, $\{a, b\} \notin E$ and $a \not\prec_V b$ as well as $b \not\prec_V a$. Thus, we can assume $|w|_a = |w|_b$. Since $h_{a,b}$ is a morphism, if w' is a prefix of w, then $h_{a,b}(w')$ is a prefix of $h_{a,b}(w)$. Therefore, $|w'|_a = |h_{a,b}(w')|_0$ and $|w'|_b = |h_{a,b}(w')|_1$. By definition, $a \not\prec_V b$ and $b \not\prec_V a$ iff there are prefixes w', w'' such that $|h_{a,b}(w')|_1 = |w'|_b < |w'|_a = |h_{a,b}(w')|_0$ and $|h_{a,b}(w'')|_0 = |w''|_a < |w''|_b = |h_{a,b}(w'')|_1$. Thus, $\{a, b\} \notin E$. Conversely, assume $\{a, b\} \notin E$. Hence, there are prefixes h', h'' of $h_{a,b}(w)$ with $|h'|_1 < |h'|_0$ and $|h''|_0 < |h''|_1$. Since $|h_{a,b}(v)| \leq 1$ for all $v \in V$, we can show by an inductive argument that, for each prefix h of $h_{a,b}(w)$, there exists a prefix w' of w such that $h = h_{a,b}(w')$. Again by definition, this means that $a \not\prec_V b$ and $b \not\prec_V a$. □

There is a rather unexpected connection between the number d of pairs of parentheses and the order dimension, a famous notion from order theory: Dyck words with d of pairs of parentheses describe exactly the comparability graphs of partial orders of dimension d, as shown in [6].

With Lemma 3, we get a characterization of co-comparability graphs in terms of complements of the symmetric hull of the Dyck language. We finish this paragraph with the symmetric hull of probably the most famous Dyck language $L_1 := \langle\{0^n1^n \mid n \in \mathbb{N}\}\rangle$ and the non-regular language $L_2 := \{w \in \{0, 1\}^* \mid |w|_0 = |w|_1\}$ where the Dyck-property is dropped (i.e., we have 2-uniform words).

Proposition 28. $(*)$ *$\mathcal{G}_{L_1}$ is the class of graph unions of co-interval graphs.*[1] *$\mathcal{G}_{L_2}$ is the class of cluster graphs.*

By Lemma 3, we get a characterization of complete multipartite graphs (as complements of cluster graphs) as $\mathcal{G}_{\overline{L_2}}$.

[1] Co-interval graphs are not closed under graph union : $2K_2$ is not a co-interval graph.

Size of the Graph Representations. The languages that we just discussed, most of which enable us to describe any graph, are not the first ones observed: for instance, $L_{\overline{1^3}}$ is such a language, as described in [11], see our discussions above. However, there is one drawback of such an encoding: the word $w(G)$ describing a given graph $G = (V, E)$ will have length $\Theta(|V|!)$, so that we need an exponential number of bits to describe a graph, measured in its order. Even worse is the situation with 1112-avoidance, see [10, Theorem 3.12], where the recursively constructed words can have length $\mathcal{O}(2^{|V|^2})$. With 2-11-representable graphs, a more concise representation of every graph was proven in [3, Theorem 5.1], using only $\mathcal{O}(n^3 \log(n))$ many bits. This is still worse than using traditional adjacency lists or adjacency matrices. In particular, the latter data structure needs $\mathcal{O}(n^2)$ bits to describe any (un)directed graph of order n. The good news is that the graph representations that we obtain with the help of palindromes, copy-words or Lyndon words are better. Taking as an example the cycle C_4, palindromes and copy languages performs much better, as illustrated in Fig. 1.

Corollary 29. *With the help of palindromes, copy-words or Lyndon words, each graph of order n can be represented with $\mathcal{O}(n^2 \log(n))$ many bits. By Lemma 3, similar statements are true for complement languages.*

Keep in mind that the log-factor has to be added, as writing a single letter of the alphabet V needs $\mathcal{O}(\log(|V|))$ many bits. Notice that this matches other implicit graph representations, e.g., in terms of sum graphs, see [8], or also that of adjacency lists. For sparse graphs, it is also important to take the number of edges into account. For instance, for adjacency lists, one needs $\mathcal{O}((n+m)\log n)$ many bits. Next, we show that a slight modification of our constructions yields a graph representation that matches this bound.

Theorem 30. $(*)$ *There exists a language L such that any graph $G = (V, E)$, with $|V| = n$ and $|E| = m$, can be represented as $G(L, w)$ for a word $w \in V^*$ of length $\mathcal{O}(n + m)$, so that w can be encoded as a binary string of size $\mathcal{O}((n+m)\log n)$.*

This theorem is also interesting from a language-theoretic perspective. While the copy-language is not context-free, its complement is even a 1-counter language. Hence, there is no need to go very far in the Chomsky hierarchy to obtain such an efficient graph encoding. On the other hand, we neither have any arguments why regular languages have to lead to essentially bigger graph encodings. We leave this as an open question.

In contrast, it might be interesting to have encodings in hand that are suitable for storing dense graphs, as opposed to sparse graphs as in Theorem 30. By the proof of this theorem, the language of copy-words $L_{\mathcal{C}}$ seems to be very suitable. For a concrete example, reconsider Corollary 14: for a vertex set V of size n, ww encodes K_n if w enumerates V in some order.

4 Conclusions and Further Open Questions

In this paper, we have shown various examples of rather famous languages that can represent large classes of graphs, if not all graphs. For instance, we showed that both Lyndon words and palindromes are suitable to represent all graphs. This motivates the question what graph class Christoffel words (introduced in [4]) can represent. Notice that they are special Lyndon words and that for a Christoffel word $w = 0v1$, v is a palindrome.

One of the nice combinatorial properties of classical word representability is that each word-representable graph can be represented by a uniform word. This kind of property does not hold for the language representations that we studied in this paper. However, there are indeed languages that can represent all graphs and that allow for uniform word representations. In [6], it is shown that every graph of so-called boxicity b can be represented by a finite $2b$-uniform language L_b, so that their union $L_B \coloneqq \bigcup_{b \geq 1} L_b$ represents all graphs as every graph has some finite boxicity [17]. However, L_B is no "nice" and well-known formal language, so that the question remains to find a "known" formal language representing all graphs by uniform words.

References

1. Alecu, B., Alekseev, V.E., Atminas, A., Lozin, V.V., Zamaraev, V.: Graph parameters, implicit representations and factorial properties. Discrete Mathematics **346**(10), 113573 (2023)
2. Berstel, J.: Transductions and Context-Free Languages, LAMM, vol. 38. Stuttgart: Teubner (1979)
3. Cheon, G.S., Kim, J., Kim, M., Kitaev, S., Pyatkin, A.: On k-11-representable graphs. Journal of Combinatorics **10**(3), 491–513 (2019)
4. Christoffel, E.B.: Observatio arithmetica. Annali di Matematica Pura ed Applicata **6**, 148–152 (1875)
5. Diekert, V., Fernau, H., Wolf, P.: Properties of graphs specified by a regular language. Acta Informatica **59**(4), 357–385 (2022)
6. Feng, Z., Fernau, H., Fleischmann, P., Kindermann, P., Sacher, S.C.: Determining factorial speed fast. Tech. Rep. 2602.24064, ArXiv, Cornell University (Feb 2026)
7. Feng, Z., Fernau, H., Mann, K., Raman, I., Sacher, S.C.: Generalized lettericity of graphs. In: Gaur, D.R., Mathew, R. (eds.) Algorithms and Discrete Applied Mathematics - 11th International Conference, CALDAM. LNCS, vol. 15536, pp. 134–146. Springer (2025)
8. Fernau, H., Gajjar, K.: The space complexity of sum labelling. Theory of Computing Systems **67**(5), 1026–1049 (2023)
9. Fleischmann, P., Haschke, L., Löck, T., Nowotka, D.: Word-representable graphs from a word's perspective. Acta Informatica **61**(4), 383–400 (2024)
10. Gaetz, M.R., Ji, C.: Enumeration and extensions of word-representants. Discrete Applied Mathematics **284**, 423–433 (2020)
11. Jones, M.E., Kitaev, S., Pyatkin, A.V., Remmel, J.B.: Representing graphs via pattern avoiding words. The Electronic Journal of Combinatorics **22**(2), (2015). P2.53

12. Kitaev, S., Lozin, V.V.: Words and Graphs. Monographs in Theoretical Computer Science. An EATCS Series, Springer (2015)
13. Kitaev, S., Pyatkin, A.V.: On representable graphs. Journal of Automata, Languages and Combinatorics **13**(1), 45–54 (2008)
14. Kitaev, S., Seif, S.: Word problem of the Perkins semigroup via directed acyclic graphs. Order **25**, 177–194 (2008)
15. Lozin, V.V.: Graph representation functions computable by finite automata. Journal of Automata, Languages and Combinatorics **13**(1), 73–90 (2008)
16. Petkovšek, M.: Letter graphs and well-quasi-order by induced subgraphs. Discrete Mathematics **244**(1), 375–388 (2002)
17. Roberts, F.S.: On the boxicity and cubicity of a graph. In: Recent Progress in Combinatorics, pp. 301–310. Academic Press (1969)